CS 59300 Application of Deep Learning

Homework 6: Machine Learning and Artificial Neural Networks

Due on **October 15**, 2025, 11:59pm

The goal of this homework assignment is to help you get familiar with machine learning models and artificial neural networks (ANN). Please keep all running outputs in all Jupyter Notebooks.

1. **XGBoost Model for Predicting Median Housing Price**. Download the class example "end_to_end_project.ipynb" file from Brightspace, rename it to "part1.ipynb", and upload it to Colab. At the end of the Jupyter Notebook, please add the XGBoost model with 100 trees and 42 for "random_state". Show the results for calling "model_cross_val_scores" function. After training the XGBoost model using the entire prepared training dataset, please show the RMSE for predicting the prepared test dataset.

2. **Avoid Overfitting in ANN**. In this part, two different technologies are applied to the class example "binary_classification_example.ipynb" in order to reduce the overfitting. The first technology is called dropout, which was developed by Geoff Hinton and his students at the University of Toronto. Dropout is one of the most effective and most commonly used regularization techniques for neural networks. Dropout, applied to a layer, consists of randomly dropping out (setting to zero) a number of output features of the layer during training. Specifically, add the following dropout layer right after each hidden layer (of two hidden layers):

   ```
   layers.Dropout(0.5)
   ```

   The second technology is called early stopping. That is, instead of finding the right epochs and then retrain the model, we can apply callbacks for the fit() function. Remove the section of "Refine the model". In the section of "Fit the model with a validation set", before training the model, add the following code for the list of callbacks:

   ```
   callbacks_list = [
      keras.callbacks.EarlyStopping(
         monitor="val_accuracy",
         patience=2,
      ),
      keras.callbacks.ModelCheckpoint(
         filepath="checkpoint_path.keras",
         monitor="val_loss",
         save_best_only=True,
      )
   ]
   ```

There are two callbacks applied: (1) Based on the validation accuracy, if it is not improved for two epochs, training will be stopped. (2) Based on the validation loss, save the best model to file “checkpoint_path.keras”.

In the “model.fit” function, add the following:

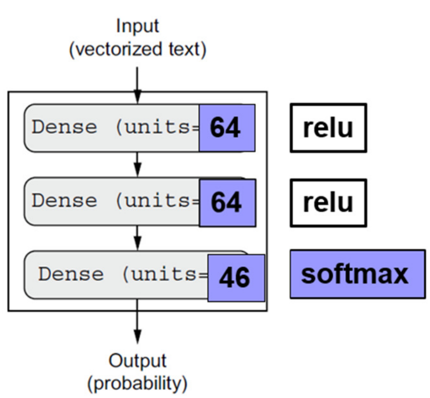
callbacks=callbacks_list,

Additionally, at the beginning of the section of “Predict new data”, add the following code to load the best model:

model = keras.models.load_model("checkpoint_path.keras")

Please run the modified code to see if the training will be automatically early stopped and the performance of prediction can be (slightly) improved due to the dropout. Name the notebook as “part2.ipynb”.

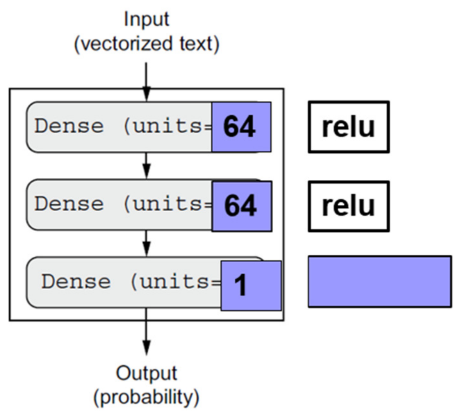
3. **Multiclass Classification in ANN**.
The goal of this problem is to classify Reuters newswires into 46 mutually exclusive topics. The Reuters dataset is a set of short newswires and their topics, published by Reuters in 1986 and has been widely used toy dataset for text classification. This dataset contains different topics and has 8,982 training examples and 2,246 test examples. It is noted that some topics are more represented than others, but each topic has at least 10 examples in the training set.
Please download part3.ipynb notebook from Brightspace and upload it to Google Colab. Fill in the missing code (8 Places). In the code, please apply the multi-hot encoding for the raw input data and the one-hot encoding for output data. Moreover, please apply the following deep learning neural network model structure:



Additionally, during the training, please apply the categorical_crossentropy loss, the rmsprop for the optimizer, the accuracy for the metrics, 20 epochs, and a batch size of 512. After finding the best epoch based on the validation loss, please retrain the model.

4. **Regression in ANN**. The goal of this problem is to predict the median price of homes in a given Boston suburb in the mid-1970s, given data points about the suburb at the time, such as the crime rate, the local property tax rate, and so on. The Boston housing dataset contains only 404 training samples and 102 test samples. Each input feature has a different scale.

   Please download part4.ipynb notebook from Brightspace and upload it to Google Colab. Fill in the missing code (5 Places). In the code, please apply the standardization method for feature scaling for the input features. Moreover, please apply the following deep learning neural network model structure:



   Additionally, during the training, please apply the mean squared error (MSE) loss, the rmsprop for the optimizer, the mean absolute error (MAE) for the metrics, 500 epochs, and a batch size of 16. After finding the best epoch based on the k-fold cross-validation MAE metric, please retrain the model.

Please upload "part1.ipynb", "part2.ipynb", "part3.ipynb", and "part4.ipynb" files to your GitHub homework repo under "hw6".

Grading rubric:

- Part 1 – 25pts
- Part 2 – 25pts
- Part 3 – 25pts
- Part 4 – 25pts

Total: 100pts